

50) The truth value of the conjunction of two propositions in fuzzy logic is the minimum of the truth values of the two propositions. What are the truth values of the statements "Fred and John are happy" and "Neither Fred nor John is happy"?

Soln: Fred and John are happy : $\min(0.8, 0.4) = 0.4$
Neither Fred nor John is happy : $\min(0.2, 0.6) = 0.2$

51) The truth value of the disjunction of two propositions in fuzzy logic is the maximum of the truth values of the two propositions. What are the truth values of the statements "Fred is happy, or John is happy" and "Fred is not happy, or John is not happy"?

Soln: Fred is happy, or John is happy : $\max(0.8, 0.4) = 0.8$
Fred is not happy, or John is not happy : $\max(0.2, 0.6) = 0.6$

52) Is the assertion "This statement is false" a proposition?

Soln: Suppose, "This statement is false" is a proposition. By defn, it can either be true or false but not both.
Suppose, the assertion is true. $\therefore$ It is true that this statement is false.
This is a ~~contradiction~~ contradiction, since a proposition can't have two opposing truth values at the same time.
Suppose, the assertion is false. $\therefore$ "This statement is false" is false. $\therefore$
The given statement is true. This is a contradiction (Same reason as above)
 $\therefore$ The assertion is not a proposition.

53) The n th statement in a list of 100 statements is "Exactly n of the statements in this list are false."

- a) What conclusions can you draw from these statements?
b) Answer part(a) if the n th statement is "At least n of the statements in this list are false."
c) Answer part(b) assuming that the list contains 99 statements.